

18-661 Introduction to Machine Learning

Support Vector Machines (SVM) – I

Spring 2024

Announcements and Reminders

- Homework 2 due today!
- Homework 3 is posted and due Feb 23.

This Week: Support Vector Machines (SVM)

Today:

- (Linear) Support Vector Machines
- Max-margin and hinge loss formulations

Next Class:

- Duality
- Kernel Machines and the “Kernel Trick”

1. Why SVM?
2. Max-Margin Formulation
3. Hinge Loss Formulation
4. Summary

Why SVM?

Why Do We Need SVM?

Naïve Bayes (circa 1750):

- $\Pr(Y|x_1 \dots x_n) = \Pr(Y) \prod_{i=1}^n \Pr(x_i|Y)$

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- $\arg \min_{\mathbf{w}} - \sum_i (y_i \log \sigma(\mathbf{w}^T \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i)))$

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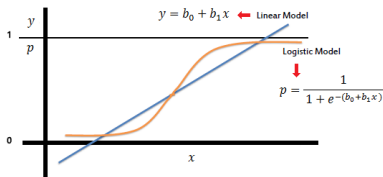
- $\arg \min_{\mathbf{w}} - \sum_i (y_i \log \sigma(\mathbf{w}^T \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i)))$

The problem (circa 1990):

- Logistic regression and Naïve Bayes train over the whole dataset.
- These can require a lot of memory in high-dimensional settings.
- Neither can be easily generalized to nonlinear settings.

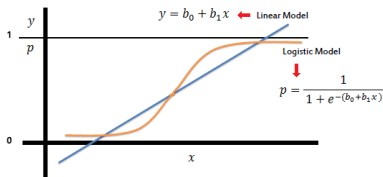
Can we do better?

Binary Logistic Regression



- We only need to know if $p(\mathbf{x}) > 0.5$ or < 0.5 .
- We **don't** (always) need to know how far $\mathbf{x}$ is from this boundary.

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How can we use this insight to improve the classification algorithm?

- What if we just looked at the boundary?
- Maybe then we could ignore some of the samples?

SVM: Support Vector Machines

Work done by researchers at AT&T Bell Labs in the 1990s.

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- Still extremely popular today

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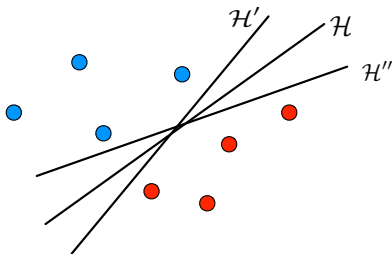
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We will see that SVM:

- Is less sensitive to outliers.
- Maximizes distance of training data from the boundary.
- Only requires a subset of the training points.
- Generalizes well to many nonlinear models.
- Scales better with high-dimensional data.

Max-Margin Formulation

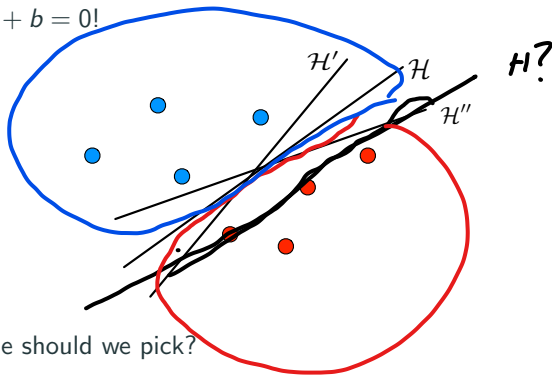
Binary Classification: Finding a Linear Decision Boundary



- Input features $\mathbf{x}$.
- Decision boundary is a hyperplane $\mathcal{H} : \mathbf{w}^\top \mathbf{x} + b = 0$.

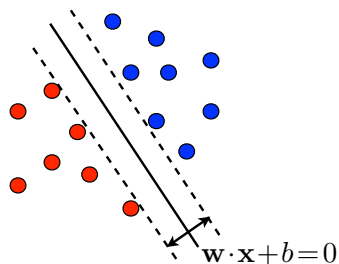
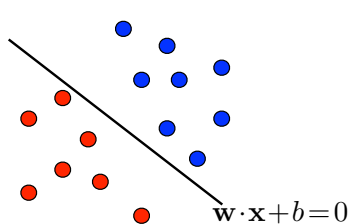
Intuition: Where to Put the Decision Boundary?

- Consider a *separable* training dataset (e.g., with two features)
- There are an **infinite** number of decision boundaries
 $\mathcal{H} : \mathbf{w}^\top \mathbf{x} + b = 0$!

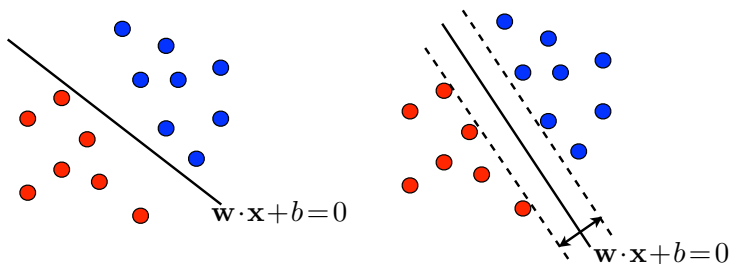


- Which one should we pick?

Intuition: Where to Put the Decision Boundary?



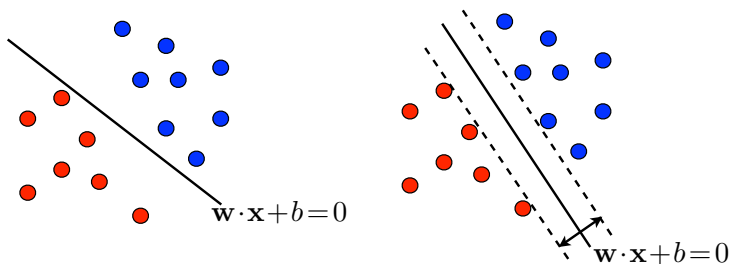
Intuition: Where to Put the Decision Boundary?



Find a decision boundary in the '*middle*' of the two classes that:

- Perfectly classifies the training data
- Is as far away from every training point as possible

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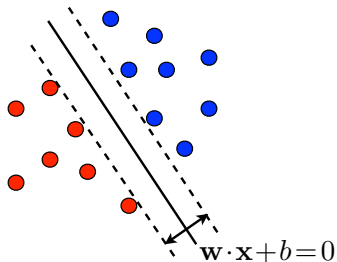
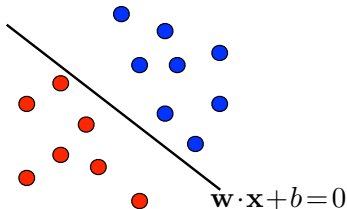
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Let us apply this intuition to build a classifier that **maximizes the margin** between training points and the decision boundary.

First, Some Vector Geometry

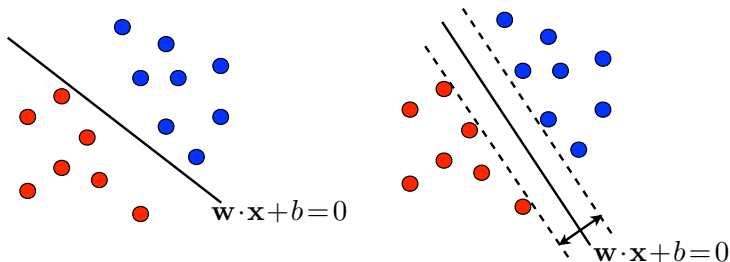
What is a hyperplane?



- General equation is $\mathbf{w}^\top \mathbf{x} + b = 0$
- Divides the space in half, i.e., $\mathbf{w}^\top \mathbf{x} + b > 0$ and $\mathbf{w}^\top \mathbf{x} + b < 0$

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- A hyperplane is a line in 2D and a plane in 3D
- $\mathbf{w} \in \mathbb{R}^d$ is a non-zero normal vector

Vector Norms and Inner Products

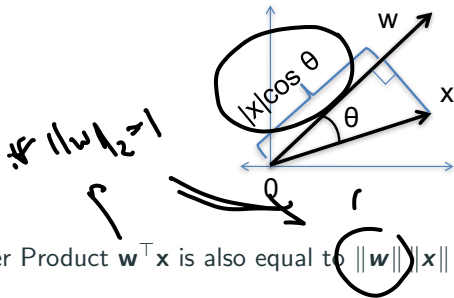
Given two vectors $\mathbf{w}$ and $\mathbf{x}$, what is their inner product?

- Inner Product $\mathbf{w}^\top \mathbf{x} = w_1x_1 + w_2x_2 + \cdots + w_dx_d$

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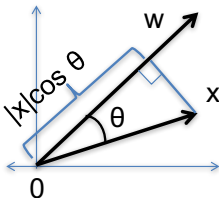


Inner Product $\mathbf{w}^\top \mathbf{x}$ is also equal to $\|\mathbf{w}\| \|\mathbf{x}\| \cos\theta$.

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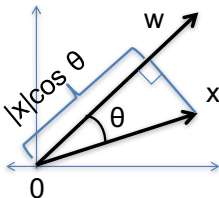
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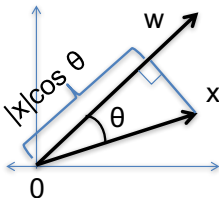
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- If $\mathbf{w} = \mathbf{x}$? $\theta = 0$, so $\mathbf{w}^\top \mathbf{w} = \|\mathbf{w}\|^2$.
- If $\mathbf{w} \perp \mathbf{x}$?

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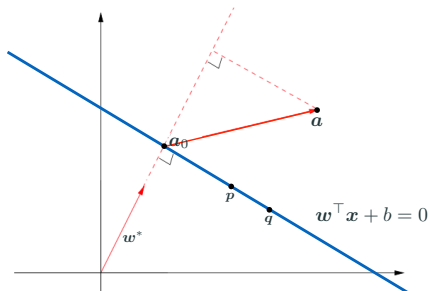
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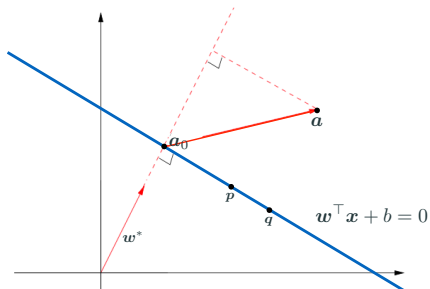
- If $\mathbf{w} = \mathbf{x}$? $\theta = 0$, so $\mathbf{w}^\top \mathbf{w} = \|\mathbf{w}\|^2$.
- If $\mathbf{w} \perp \mathbf{x}$? $\theta = \pi/2$, so $\mathbf{w}^\top \mathbf{x} = 0$.

Normal Vector of a Hyperplane



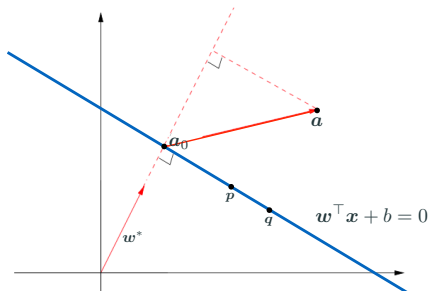
What is the meaning of w in the hyperplane $w^T x + b = 0$?

Normal Vector of a Hyperplane



Vector w is normal to the hyperplane. Why?

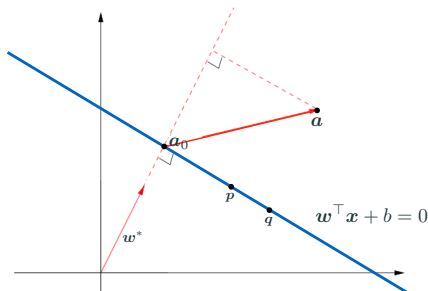
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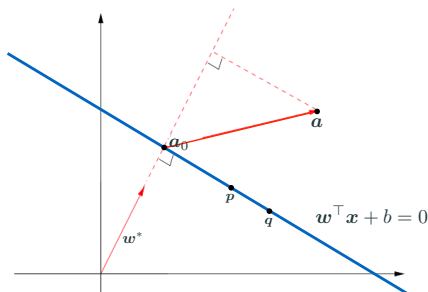
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- Then $w^\top (p - q) = w^\top p - w^\top q = -b - (-b) = 0$

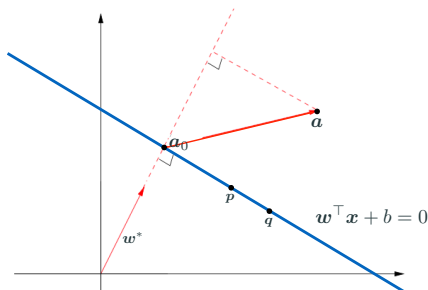
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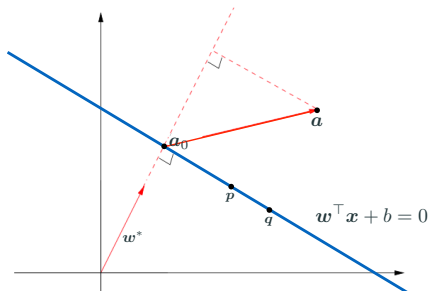


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Let $w^* = \frac{w}{\|w\|_2}$ be the unit normal vector in the direction w .

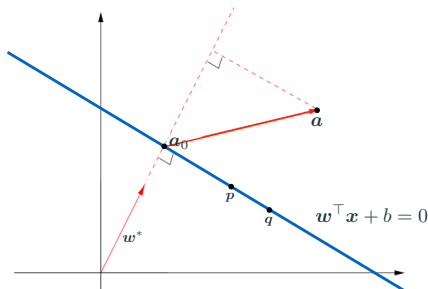
Distance from a Hyperplane



How to find the distance from a to the hyperplane?

- We want to find distance between a and line in the direction of w^* .

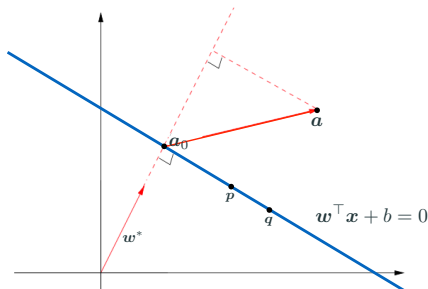
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- We want to find distance between $\mathbf{a}$ and line in the direction of $\mathbf{w}^*$.
- If we define point $\mathbf{a}_0$ on the line, then this distance corresponds to length of $\mathbf{a} - \mathbf{a}_0$ in direction of $\mathbf{w}^*$, which equals $\mathbf{w}^{*\top}(\mathbf{a} - \mathbf{a}_0)$.

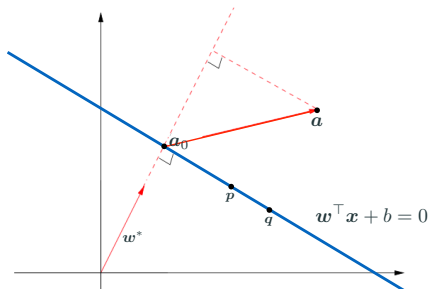
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- We know $w^\top a_0 = -b$ since $w^\top a_0 + b = 0$.
- Then the distance equals $\frac{1}{\|w\|_2}(w^\top a + b)$.

Distance from a Point to Decision Boundary

The *unsigned* distance from a point $\mathbf{x}$ to the decision boundary (hyperplane) $\mathcal{H}$ is

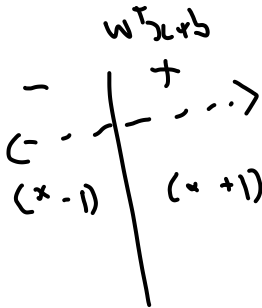
$$d_{\mathcal{H}}(\mathbf{x}) = \frac{|\mathbf{w}^{\top} \mathbf{x} + b|}{\|\mathbf{w}\|_2}$$

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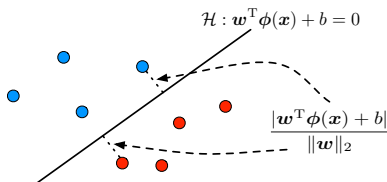
Then, exploiting the fact that the decision boundary classifies every point in the training dataset correctly, we have $(\mathbf{w}^{\top} \mathbf{x} + b)$ and $\mathbf{x}$'s label y must have the same sign. So we get

$$d_{\mathcal{H}}(\mathbf{x}) = \frac{y[\mathbf{w}^{\top} \mathbf{x} + b]}{\|\mathbf{w}\|_2}$$

Defining the Margin

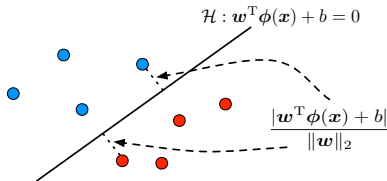
Margin: Smallest distance between the hyperplane and all training points

$$\text{MARGIN}(\mathbf{w}, b) = \min_n \frac{y_n[\mathbf{w}^\top \mathbf{x}_n + b]}{\|\mathbf{w}\|_2}$$



How can we use this to find the SVM solution?

Optimizing the Margin

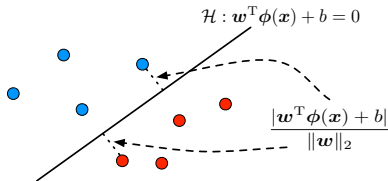


How should we pick $(\mathbf{w}, b)$ based on its margin?

We want a decision boundary that is as far away from all training points as possible, so we to *maximize* the margin!

$$\max_{\mathbf{w}, b} \left(\min_n \frac{y_n [\mathbf{w}^T \mathbf{x}_n + b]}{\|\mathbf{w}\|_2} \right) = \max_{\mathbf{w}, b} \left(\frac{1}{\|\mathbf{w}\|_2} \min_n y_n [\mathbf{w}^T \mathbf{x}_n + b] \right)$$

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Only involves points near the boundary (more on this later).

Margin: Smallest distance between the hyperplane and all training points

$$\text{MARGIN}(\mathbf{w}, b) = \min_n \frac{y_n[\mathbf{w}^\top \mathbf{x}_n + b]}{\|\mathbf{w}\|_2}$$

Consider three hyperplanes

$$(\mathbf{w}, b) \quad (2\mathbf{w}, 2b) \quad (.5\mathbf{w}, .5b)$$

Which one has the largest margin?

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Which one has the largest margin?

- The MARGIN doesn't change if we scale $(\mathbf{w}, b)$ by a constant c
- $\mathbf{w}^\top \mathbf{x} + b = 0$ and $(c\mathbf{w})^\top \mathbf{x} + (cb) = 0$: same decision boundary!

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- $\mathbf{w}^\top \mathbf{x} + b = 0$ and $(c\mathbf{w})^\top \mathbf{x} + (cb) = 0$: same decision boundary!
- Can we further constrain the problem so as to get a unique solution $(\mathbf{w}, b)$?

Rescaled Margin

We can further constrain the problem by scaling $(\mathbf{w}, b)$ such that

$$\min_n y_n [\mathbf{w}^\top \mathbf{x}_n + b] = 1.$$

Note that there always exists a scaling for which this is true.

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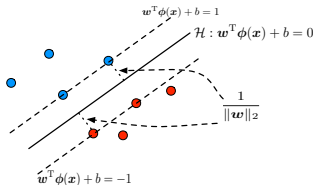
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Hence the points closest to the decision boundary are at distance $\frac{1}{\|\mathbf{w}\|_2}$.

$$\mathbf{w}^\top \mathbf{x} + b = 0$$

$$(\mathbf{w} \cos)^\top \mathbf{x} + b \cos = 0$$



SVM: Max-margin Formulation for Separable Data

We thus want to solve:

$$\max_{\mathbf{w}, b} \underbrace{\frac{1}{\|\mathbf{w}\|_2}}_{\text{margin}} \quad \text{such that} \quad \underbrace{\min_n y_n [\mathbf{w}^\top \mathbf{x}_n + b]}_{\text{scaling of } \mathbf{w}, b} = 1$$

which is equivalent to

$$\max_{\mathbf{w}, b} \frac{1}{\|\mathbf{w}\|_2} \quad \text{such that} \quad y_n [\mathbf{w}^\top \mathbf{x}_n + b] \geq 1, \quad \forall n$$

$$\text{If } \min_n \beta_n = 1 \iff \forall n \beta_n \geq 1$$

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This is further equivalent to

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y_n [\mathbf{w}^\top \mathbf{x}_n + b] \geq 1, \quad \forall n \end{aligned}$$

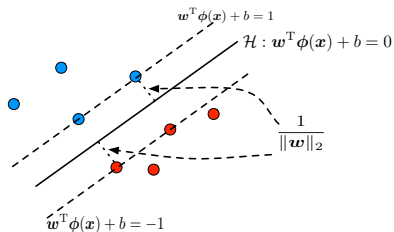
Given our geometric intuition, SVM is called a **max margin** (or large margin) classifier. The constraints are called **large margin constraints**.

Support Vectors: A First Look

SVM formulation for separable data

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} \|\mathbf{w}\|_2^2$$

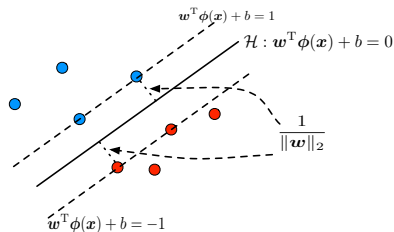
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Two types of training data, based on the situations of the constraint:

- “=”: $y_n[\mathbf{w}^\top \mathbf{x}_n + b] = 1$. These training data points are called “support vectors”, which have the minimum distance $(\frac{1}{\|\mathbf{w}\|})$ to the boundary.

Support Vectors: A First Look

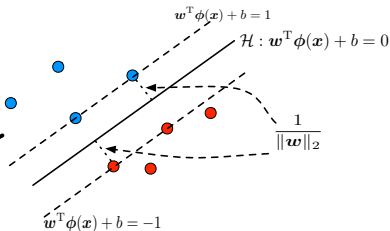
SVM formulation for separable data

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} \|\mathbf{w}\|_2^2$$

$$\text{s.t. } y_n[\mathbf{w}^\top \mathbf{x}_n + b] \geq 1, \quad \forall n$$

Constraints

$\longleftrightarrow$ max margin



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- “=”: $y_n[\mathbf{w}^\top \mathbf{x}_n + b] = 1$. These training data points are called “support vectors”, which have the minimum distance $(\frac{1}{\|\mathbf{w}\|})$ to the boundary.
- “>”: $y_n[\mathbf{w}^\top \mathbf{x}_n + b] > 1$. Distance to the boundary is larger than the minimum. Removing these data points does not affect the optimal solution (more on this next lecture).

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Non-separable setting

In practice our training data may not be separable. What issues arise with the optimization problem above when data is not separable?

SVM for Non-separable Data

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- For every $\mathbf{w}$ there exists a training point $\mathbf{x}_i$ such that

$$y_i[\mathbf{w}^\top \mathbf{x}_i + b] \leq 0$$

- There is no feasible $(\mathbf{w}, b)$ as at least one of our constraints is violated!

Constraints in separable setting

$$y_n[\mathbf{w}^\top \mathbf{x}_n + b] \geq 1, \quad \forall n$$

Constraints in non-separable setting

Can we modify our constraints to account for non-separability?

SVM for Non-separable Data

Constraints in separable setting

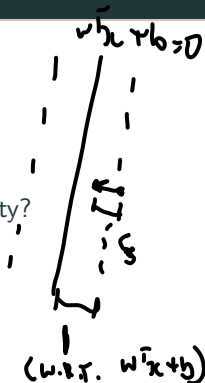
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Specifically, we introduce **slack variables** $\xi_n \geq 0$:

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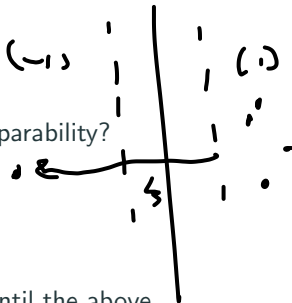
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- For “hard” training points, we can increase ξ_n until the above inequalities are met.
- What does it mean when $\xi_n = 0$? This data point is correctly classified.
- What does it mean when ξ_n is very large? We have violated the original constraints “by a lot.”

Soft-margin SVM Formulation

We do not want ξ_n to grow too large, and we can control their size by incorporating them into our optimization problem:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_n \xi_n \\ \text{s.t.} \quad & y_n [\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n, \quad \forall n \\ & \xi_n \geq 0, \quad \forall n \end{aligned}$$

Handwritten notes: An arrow points from the text "we pick" to the parameter C . A horizontal line is drawn under the constraint $y_n [\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n$.

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What is the role of C ?

- User-defined hyperparameter
- Trades off between the two terms in our objective
- Same idea as the regularization term in ridge regression

How to Solve this Problem?

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_n \xi_n \\ \text{s.t.} \quad & y_n [\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n, \quad \forall n \\ & \xi_n \geq 0, \quad \forall n \end{aligned}$$

- This is a **convex quadratic program**: the objective function is quadratic in $\mathbf{w}$ and linear in ξ and the constraints are linear (inequality) constraints in $\mathbf{w}$, b and ξ_n .

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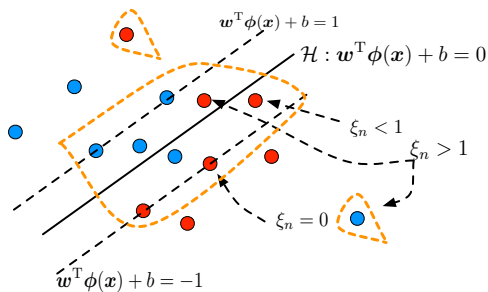
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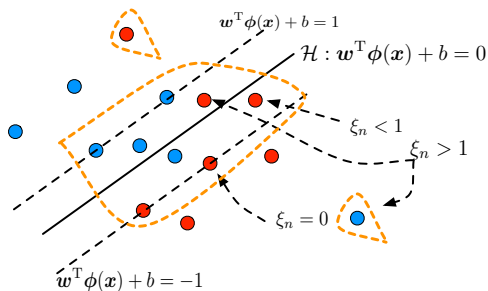
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- Early solvers were based on general-purpose quadratic program solvers (e.g. similar to `scipy.optimize` or Matlab's `quadprog()`, albeit in the 1990s)
- SVM solvers today are based on highly optimized search algorithms that exploit SVM-specific structure.

Support Vectors: Revisited



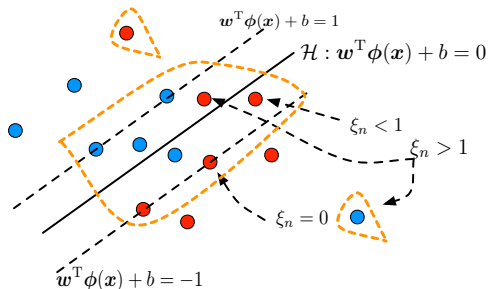
Support vectors are highlighted by the dotted orange lines. What does this mean mathematically?

Support Vectors: Revisited



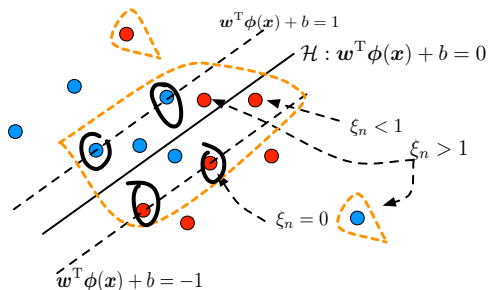
Recall the constraints $y_n[\mathbf{w}^T \mathbf{x}_n + b] \geq 1 - \xi_n$ from the soft-margin formulation. All the training points $(\mathbf{x}_n, y_n)$ that satisfies the constraint with “=” are support vectors.

Support Vectors: Revisited



In other words, support vectors satisfy $y_n[\mathbf{w}^T \mathbf{x}_n + b] = 1 - \xi_n$, which can be further divided into several categories:

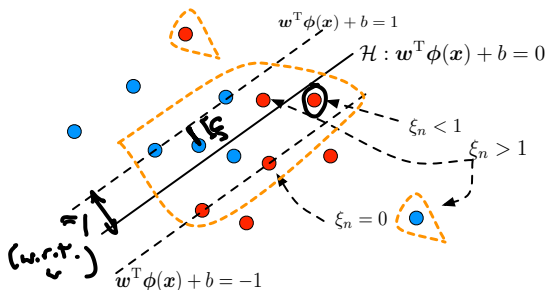
Support Vectors: Revisited



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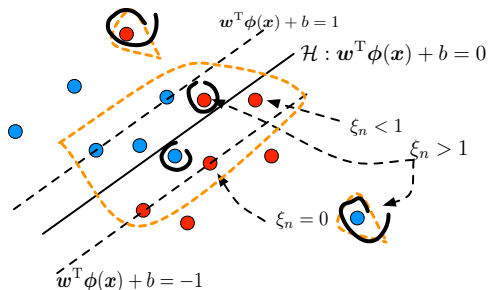
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- $0 < \xi_n \leq 1$: $y_n[\mathbf{w}^T \mathbf{x}_n + b] \in [0, 1)$ on the correct side, but with distance less than $\frac{1}{\|\mathbf{w}\|}$.

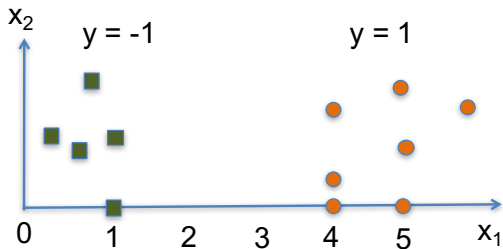
Support Vectors: Revisited



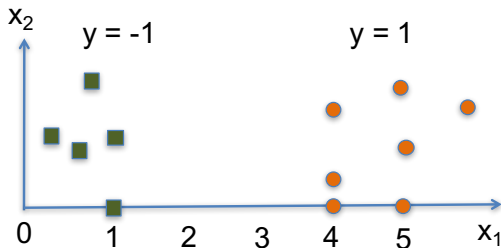
In other words, support vectors satisfy $y_n[\mathbf{w}^T \mathbf{x}_n + b] = \underline{1 - \xi_n}$, which can be further divided into several categories:

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- $\xi_n > 1$: $y_n[\mathbf{w}^T \mathbf{x}_n + b] < 0$, on the wrong side of the boundary.

Example of SVM

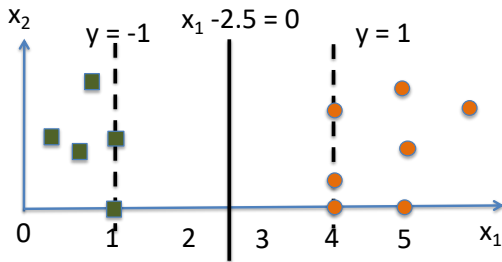


Example of SVM



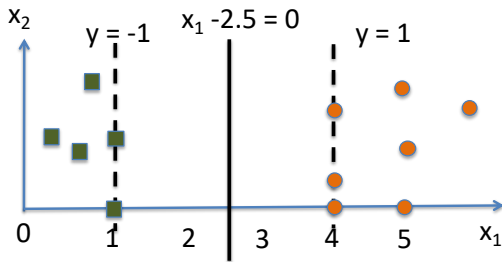
What will be the decision boundary learnt by solving the SVM optimization problem?

Example of SVM



Margin = 1.5; the decision boundary has $\mathbf{w} = [1, 0]^T$, and $b = -2.5$.

Example of SVM

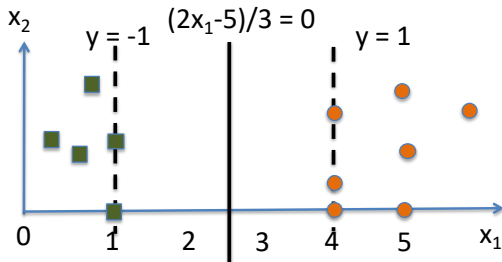


Margin = 1.5; the decision boundary has $\mathbf{w} = [1, 0]^\top$, and $b = -2.5$.

Not quite: we need the support vectors to satisfy to $y_n(\mathbf{w}^\top \mathbf{x}_n + b) = 1$.
For example, for $\mathbf{x}_n = [1, 0]^\top$, we have

$$y_n(\mathbf{w}^\top \mathbf{x}_n + b) = (-1)[1 - 2.5] = 1.5.$$

Example of SVM: Scaling



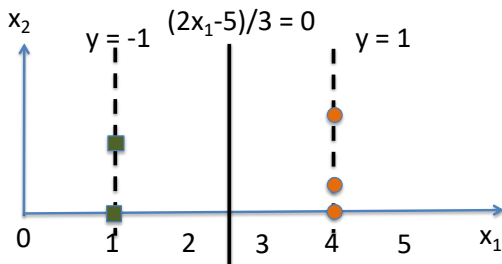
Thus, our optimization problem will re-scale $\mathbf{w}$ and b to get this equation for the same decision boundary.

Margin = 1.5; the decision boundary has $\mathbf{w} = [2/3, 0]^\top$, and $b = -5/3$.

For example, for $\mathbf{x}_n = [1, 0]^\top$, we have

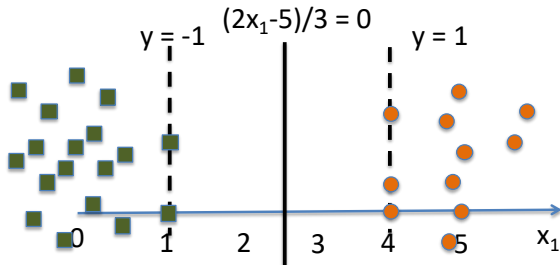
$$y_n(\mathbf{w}^\top \mathbf{x}_n + b) = (-1)[2/3 - 5/3] = 1.$$

Example of SVM: Support Vectors



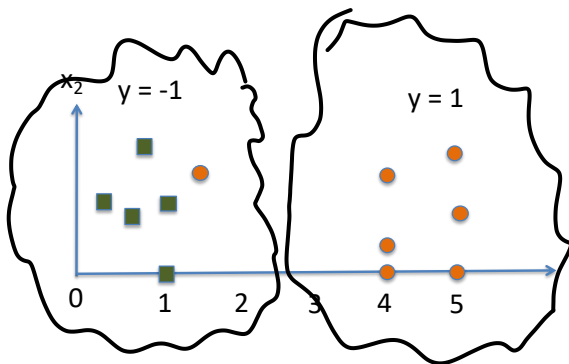
The solution to our optimization problem will be the **same** to the *reduced* dataset containing all the support vectors.

Example of SVM: Support Vectors



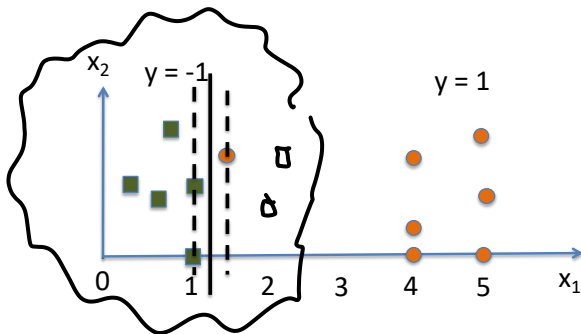
There can be many more data than the number of support vectors (so we can train on a smaller dataset).

Example of SVM: Resilience to Outliers



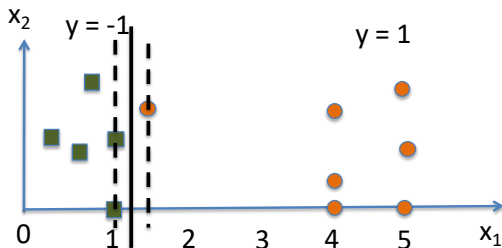
- Still linearly separable, but one of the orange dots is an “outlier”.

Example of SVM: Resilience to Outliers



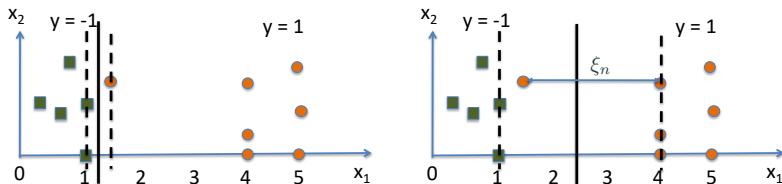
- Naively applying the hard-margin SVM will result in a classifier with small margin.

Example of SVM: Resilience to Outliers



- Naively applying the hard-margin SVM will result in a classifier with small margin.
- So, better to use the soft-margin (or equivalently, hinge loss) formulation.

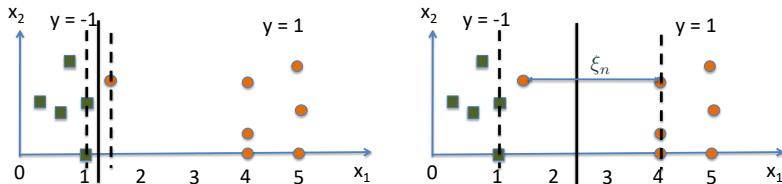
Example of SVM: Resilience to Outliers



$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_n \xi_n \\ \text{s.t.} \quad & y_n [\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n, \quad \forall n \\ & \xi_n \geq 0, \quad \forall n \end{aligned}$$

Due to the flexibility provided by C , (properly tuned) SVM is less sensitive to outliers.

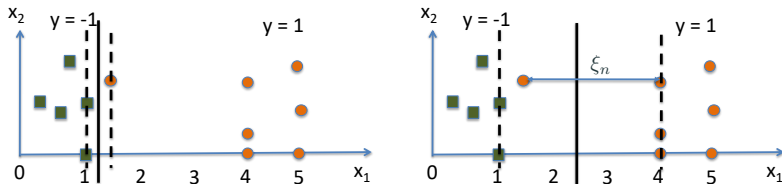
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- What happens if C is very small?

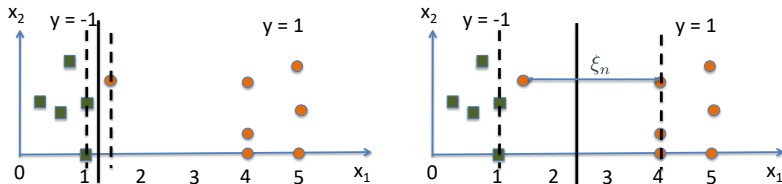
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- What happens if C is very small? More data points near the boundary are disregarded.

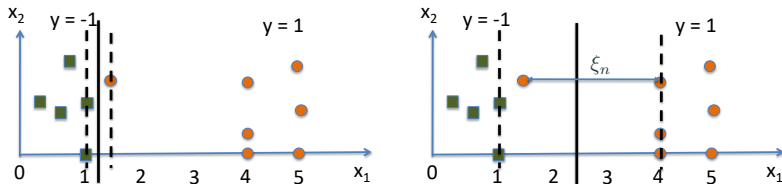
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- What happens if C is very small? More data points near the boundary are disregarded.
- What happens if C is 0?

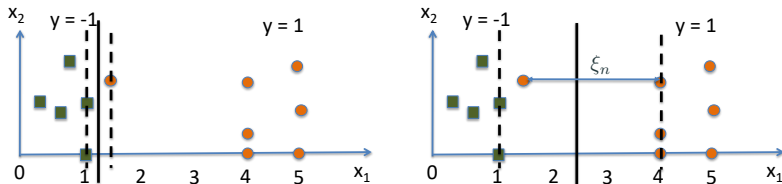
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- What happens if C is 0? All data points will be ignored.

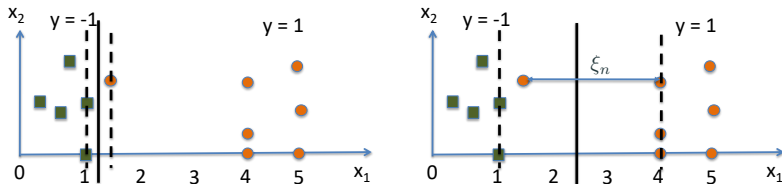
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- What happens if C is very large?

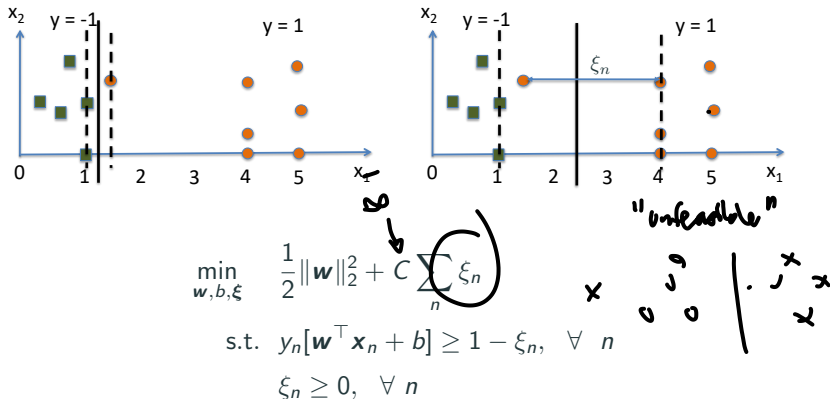
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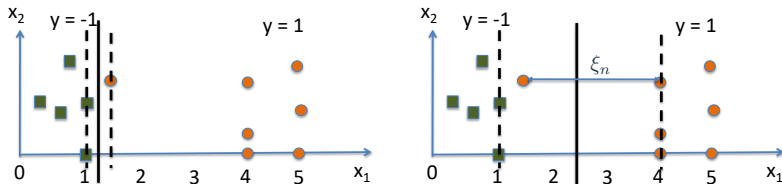
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Example of SVM: Resilience to Outliers



- What happens if C is very large? Outliers near the decision boundary will have a greater impact.
- What happens if C is ∞ ?

Example of SVM: Resilience to Outliers



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- What happens if C is very large? Outliers near the decision boundary will have a greater impact.
- What happens if C is ∞ ? We get hard margin SVM.

Hinge Loss Formulation

SVM vs. Logistic Regression

SVM soft-margin formulation

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_n \xi_n \\ \text{s.t.} \quad & y_n [\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n, \quad \forall n \\ & \xi_n \geq 0, \quad \forall n \end{aligned}$$

Logistic regression formulation

$$\begin{aligned} \min_{\mathbf{w}} \quad & - \sum_n \{y_n \log \sigma(\mathbf{w}^\top \mathbf{x}_n) \\ & + (1 - y_n) \log [1 - \sigma(\mathbf{w}^\top \mathbf{x}_n)]\} \\ & + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \end{aligned}$$

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- Logistic regression defines a **loss for each data point** and minimizes the total loss plus a regularization term.
- This is convenient for assessing the “goodness” of the model on each data point.
- Can we write SVMs in this form as well? **The Hinge Loss formulation!**

Derive the Hinge Loss Formulation

Here's the soft-margin formulation again:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_n \xi_n \quad \text{s.t.} \quad y_n[\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n, \quad \xi_n \geq 0, \quad \forall n$$

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Now since $\underline{y_n[\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n} \iff \underline{\xi_n \geq 1 - y_n[\mathbf{w}^\top \mathbf{x}_n + b]}$:

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$$\min_{\mathbf{w}, b, \xi} C \sum_n \xi_n + \frac{1}{2} \|\mathbf{w}\|_2^2 \quad \text{s.t.} \quad \xi_n \geq \max(0, 1 - y_n[\mathbf{w}^\top \mathbf{x}_n + b]), \quad \forall n$$

Derive the Hinge Loss Formulation

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Now since the ξ_n should always be as small as possible, we obtain:

$$\min_{\mathbf{w}, b} C \sum_n \max(0, 1 - y_n[\mathbf{w}^\top \mathbf{x}_n + b]) + \frac{1}{2} \|\mathbf{w}\|_2^2$$

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Divide by C and set $\lambda = \frac{1}{C}$, we get **Hinge Loss formulation**:

$$\min_{\mathbf{w}, b} \sum_n \underbrace{\max(0, 1 - y_n[\mathbf{w}^\top \mathbf{x}_n + b])}_{\text{Hinge Loss for } x_n, y_n} + \left(\frac{\lambda}{2} \|\mathbf{w}\|_2^2 \right)$$

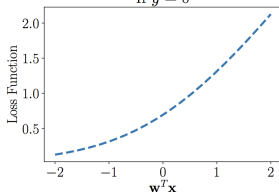
Logistic Regression Loss vs Hinge Loss

Given training data $(\mathbf{x}_n, y_n)$, the cross entropy loss was

$$-\{y_n \log \sigma(\mathbf{w}^\top \mathbf{x}_n) + (1 - y_n) \log[1 - \sigma(\mathbf{w}^\top \mathbf{x}_n)]\}$$

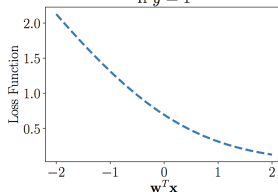
$$-\log \left(\frac{e^{-\mathbf{w}^\top \mathbf{x}_n}}{1 + e^{-\mathbf{w}^\top \mathbf{x}_n}} \right)$$

If $y = 0$



$$-\log \left(\frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}_n}} \right)$$

If $y = 1$

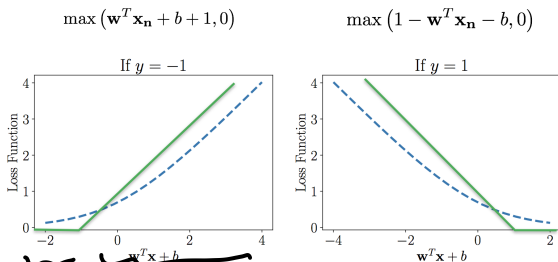


- What does the Hinge Loss Function look like?

Logistic Regression Loss vs Hinge Loss

Given training data $(\mathbf{x}_n, y_n)$, the Hinge loss is

$$\max(0, 1 - y_n[\mathbf{w}^\top \mathbf{x}_n + b])$$



*not violating
i.e. not a support vector*

- Loss grows linearly as we move away from the boundary.
- No penalty if a point is more than 1 unit from the boundary.

Hinge Loss SVM Formulation

Minimizing the total hinge loss on all the training data

$$\min_{\mathbf{w}, b} \sum_n \underbrace{\max(0, 1 - y_n[\mathbf{w}^\top \mathbf{x}_n + b])}_{\text{hinge loss for sample } n} + \underbrace{\frac{\lambda}{2} \|\mathbf{w}\|_2^2}_{\text{regularizer}}$$

Analogous to regularized least squares or logistic regression, as we balance between two terms (the loss and the regularizer).

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- Can solve using gradient descent to get the optimal $\mathbf{w}$ and b
- Gradient of the first term will be either 0, $\mathbf{x}_n$ or $-\mathbf{x}_n$ depending on y_n and $\mathbf{w}^\top \mathbf{x}_n + b$.

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Analogous to regularized least squares or logistic regression, as we balance between two terms (the loss and the regularizer).

- Can solve using gradient descent to get the optimal $\mathbf{w}$ and b
- Gradient of the first term will be either 0, $\mathbf{x}_n$ or $-\mathbf{x}_n$ depending on y_n and $\mathbf{w}^\top \mathbf{x}_n + b$.
- Much easier to compute than in logistic regression, where we need to compute the sigmoid function $\sigma(\mathbf{w}^\top \mathbf{x}_n + b)$ in each iteration.

Summary

Hard-margin (for separable data)

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|_2^2 \text{ s.t. } y_n[\mathbf{w}^\top \mathbf{x}_n + b] \geq 1, \xi_n \geq 0, \forall n$$

Three SVM Formulations

Hard-margin (for separable data)

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Soft-margin (add slack variables)

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_n \xi_n \text{ s.t. } y_n[\mathbf{w}^\top \mathbf{x}_n + b] \geq 1 - \xi_n, \xi_n \geq 0, \forall n$$

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Hinge loss (define a loss function for each data point)

$$\min_{\mathbf{w}, b} \sum_n \max(0, 1 - y_n[\mathbf{w}^\top \mathbf{x}_n + b]) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

Advantages of SVM

We've seen the geometric formulation of SVM and the equivalent formulation of minimizing the empirical hinge loss.

This explains why SVM:

1. Is less sensitive to outliers.
2. Maximizes distance of training data from the boundary.
3. Only requires a subset of the training points.
4. Generalizes well to many nonlinear models.
5. Scales better with high-dimensional data.

We will need to use **duality** to show the remaining properties.

You should know:

- Max-margin formulation for separable and non-separable SVMs.
- Definition and importance of support vectors.
- Hinge loss formulation of SVMs.
- Equivalence of the max-margin and hinge loss formulations.

Next class:

- Duality
- Nonlinear SVM